

Final Project

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Libraries

```
library(readxl)
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2     3.5.1      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr       1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

Loading Data

```
Budget_vs_Spending_Data <- read_excel("C:/Users/renzo/OneDrive/Escritorio/DATA 211-01 Data Science and V
View(Budget_vs_Spending_Data)

head(Budget_vs_Spending_Data)
```

```
## # A tibble: 6 x 3
##   Date           Daily_Budget Daily_Spending
##   <dtm>          <dbl>         <dbl>
## 1 2025-04-06 00:00:00         42          64.4
## 2 2025-04-07 00:00:00         39          61.2
## 3 2025-04-08 00:00:00         43          35.4
## 4 2025-04-09 00:00:00         47          44.8
## 5 2025-04-10 00:00:00         38          28.5
## 6 2025-04-11 00:00:00         38          33.5
```

Data Preparation

Prepare the dataset by converting the Date column to the correct type and reshaping the data for visualizations.

```
Budget_vs_Spending_Data$Date <- as.Date(Budget_vs_Spending_Data$Date, format = "%m/%d/%Y")

long_data <- pivot_longer(Budget_vs_Spending_Data,
                           cols = c(Daily_Budget, Daily_Spending),
                           names_to = "Type",
                           values_to = "Amount")

head(long_data)
```

```
## # A tibble: 6 x 3
##   Date      Type      Amount
##   <date>    <chr>    <dbl>
## 1 2025-04-06 Daily_Budget    42
## 2 2025-04-06 Daily_Spending  64.4
## 3 2025-04-07 Daily_Budget    39
## 4 2025-04-07 Daily_Spending  61.2
## 5 2025-04-08 Daily_Budget    43
## 6 2025-04-08 Daily_Spending  35.4
```

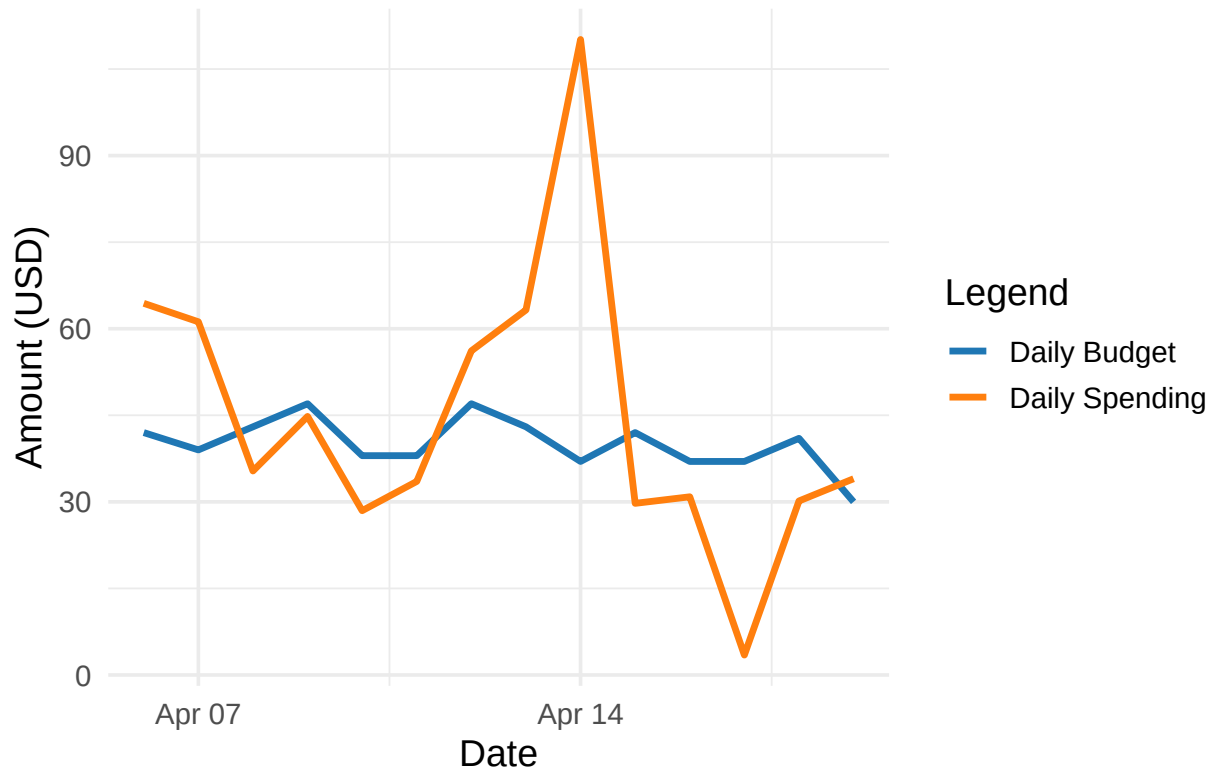
Line Plot - Daily Budget vs. Daily Spending

Visualize how your daily budget and spending change over time.

```
ggplot(Budget_vs_Spending_Data, aes(x = Date)) +
  geom_line(aes(y = Daily_Budget, color = "Daily Budget"), size = 1.2) +
  geom_line(aes(y = Daily_Spending, color = "Daily Spending"), size = 1.2) +
  scale_color_manual(values = c("Daily Budget" = "#1f77b4", "Daily Spending" = "#ff7f0e")) +
  labs(title = "Daily Budget vs. Daily Spending",
       x = "Date", y = "Amount (USD)", color = "Legend") +
  theme_minimal(base_size = 14)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Daily Budget vs. Daily Spending



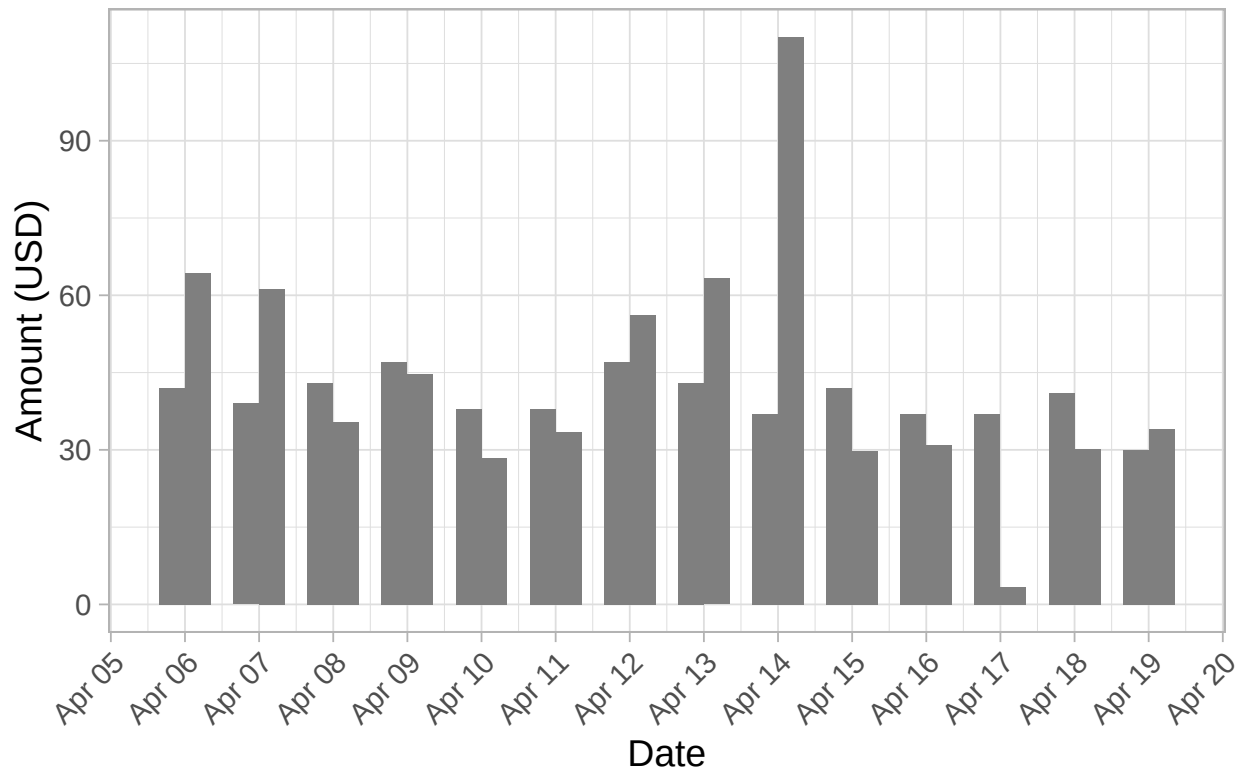
Bar Plot - Daily Comparison

Compare daily budget and daily spending side-by-side for each day.

```
ggplot(long_data, aes(x = Date, y = Amount, fill = Type)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = c("Daily Budget" = "#1f77b4", "Daily Spending" = "#ff7f0e")) +
  labs(title = "Daily Budget vs. Daily Spending (Side-by-Side)",
       x = "Date", y = "Amount (USD)", fill = "Type") +
  theme_light(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_date(date_breaks = "1 day", date_labels = "%b %d")
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's fill values.
## No shared levels found between 'names(values)' of the manual scale and the
## data's fill values.
```

Daily Budget vs. Daily Spending (Side-by-Side)



Boxplot - Distribution and Outliers

Understand the distribution and detect potential outliers in your spending and budgeting.

```
ggplot(long_data, aes(x = Type, y = Amount, fill = Type)) +  
  geom_boxplot(alpha = 0.7, outlier.color = "red", outlier.size = 3) +  
  scale_fill_manual(values = c("Daily Budget" = "#1f77b4", "Daily Spending" = "#ff7f0e")) +  
  labs(title = "Distribution of Budget and Spending",  
        x = "", y = "Amount (USD)") +  
  theme_minimal(base_size = 14)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the  
## data's fill values.  
## No shared levels found between 'names(values)' of the manual scale and the  
## data's fill values.
```

Distribution of Budget and Spending



Statistical Analysis: Paired t-Test

We want to test whether there is a significant difference between the mean Daily Budget and the mean Daily Spending.

$$H_0 : \mu_{\text{Daily Budget}} = \mu_{\text{Daily Spending}}$$

$$H_a : \mu_{\text{Daily Budget}} \neq \mu_{\text{Daily Spending}}$$

Where:

- $\mu_{\text{Daily Budget}}$ = Mean of the Daily Budget
- $\mu_{\text{Daily Spending}}$ = Mean of the Daily Spending

This is a **two-tailed paired t-test**.

```
t_test_result <- t.test(Budget_vs_Spending_Data$Daily_Budget,
                        Budget_vs_Spending_Data$Daily_Spending,
                        paired = TRUE,
                        alternative = "two.sided")
```

```
t_test_result
```

```
##
## Paired t-test
```

```
##  
## data: Budget_vs_Spending_Data$Daily_Budget and Budget_vs_Spending_Data$Daily_Spending  
## t = -0.68809, df = 13, p-value = 0.5035  
## alternative hypothesis: true mean difference is not equal to 0  
## 95 percent confidence interval:  
## -19.072084 9.857798  
## sample estimates:  
## mean difference  
## -4.607143
```

Conclusion

Since the p -value = 0.5035 is greater than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis H_0 . At the 5% significance level, we do not have enough evidence to conclude that there is a significant difference between the average Daily Budget (μ_{budget}) and the average Daily Spending (μ_{spending}).